

OWEN M. DUGAN

(914)841-0007 | odugan@stanford.edu | [linkedin.com/in/owen-m-dugan](https://www.linkedin.com/in/owen-m-dugan) | druidowm.github.io

EDUCATION

Ph.D. Computer Science, Stanford University	TBD	2024 - Present
B.S. Physics, MIT	5.0/5.0	2021 - 2024

RESEARCH INTERESTS

ML systems: kernel fusion, GPU scheduling, and high-throughput/low-latency inference.

ML architectures: using physics and math to develop more capable and efficient machine learning architectures.

SELECT AWARDS & ACCOMPLISHMENTS

Francis F. Lee Memorial Fellow, Fannie and John Hertz Foundation	2025
Hertz Fellow, Fannie and John Hertz Foundation	2024
Knight-Hennessy Scholar, Stanford University Knight-Hennessy Scholarship	2024
NSF Graduate Research Fellow, National Science Foundation	2024
Outstanding UROP (Undergraduate Research) Award, MIT School of Science	2023
U.S. Patent Nos. 11,688,045 and 11,756,304, US Patent & Trademark Office	2023
William Lowell Putnam Mathematical Competition, 164th place (top 5%)	2022
Neo Scholar, Neo Venture Capital Firm	2022
US Presidential Scholar, US Department of Education	2021
Davidson Fellows Scholar, Davidson Institute	2021
Regeneron Science Talent Search Scholar, Regeneron Science Talent Search	2021
RSI Scholar, Research Science Institute	2020
SAT, 1600 (Perfect Score)	2020
Caroline D. Bradley Scholar, Institute for Educational Advancement	2016

RELEVANT COURSEWORK

Computer science

Graduate Level: Reinforcement Learning.

Undergraduate Level: Algorithms 1 & 2, Automata Theory & Decidability, Computer Vision.

Physics

Graduate Level: Quantum Theory, Quantum Field Theory 1 & 2, Statistical Mechanics, Solid-State Physics.

Undergraduate Level: Quantum Computing, Classical Mechanics, Experimental Physics, Physics Thesis.

Mathematics

Undergraduate Level: Calculus, Multivariable Calculus, Counting and Probability 1 & 2, Number Theory 1 & 2, Linear Algebra, Differential Equations, Real Analysis, Complex Analysis, Abstract Algebra 1 & 2.

PUBLICATIONS

17) Benjamin Spector, Jordan Juravsky, Stuart Sul, Dylan Lim, **Owen Dugan**, Simran Arora, Chris Ré
We Bought the Whole GPU, So We're Damn Well Going to Use the Whole GPU (September 28, 2025)

[HAZYRESEARCH.STANFORD.EDU/BLOG/2025-09-28-TP-LLAMA-MAIN](https://hazyresearch.stanford.edu/blog/2025-09-28-tp-llama-main)

16) Benjamin Spector, Jordan Juravsky, Stuart Sul, Dylan Lim, **Owen Dugan**, Simran Arora, Chris Ré
How Many Llamas Can Dance in the Span of a Kernel? (September 28, 2025)

[HAZYRESEARCH.STANFORD.EDU/BLOG/2025-09-28-TP-LLAMA-INTRO](https://hazyresearch.stanford.edu/blog/2025-09-28-tp-llama-intro)

15) Benjamin Spector, Jordan Juravsky, Stuart Sul, **Owen Dugan**, Dylan Lim, Dan Fu, Simran Arora, Chris Ré
Look Ma, No Bubbles! Designing a Low-Latency Megakernel for Llama-1B (May 27, 2025)

[HAZYRESEARCH.STANFORD.EDU/BLOG/2025-05-27-NO-BUBBLES](https://hazyresearch.stanford.edu/blog/2025-05-27-no-bubbles)

- 14) Jerry Liu, Jessica Grogan, **Owen Dugan**, Ashish Rao, Simran Arora, Atri Rudra, Christopher Ré
Towards Learning High-Precision Least Squares Algorithms with Sequence Models
International Conference on Learning Representations (ICLR), 2025.
[ARXIV:2503.12295](#)
- 13) A. Ma, **O. Dugan**, M. Soljačić
Predicting Band Gap from Chemical Composition: A Simple Learned Model for a Material Property with Atypical Statistics
arXiv preprint [ARXIV:2501.02932](#), 2025.
- 12) **Owen Dugan**, Donato Manuel Jimenez Beneto, Charlotte Loh, Zhuo Chen, Rumen Dangovski, Marin Soljačić
OccamLLM: Fast and Exact Language Model Arithmetic in a Single Step
[ARXIV:2406.06576](#)
- 11) Zhuo Chen, Rumen Dangovski, Charlotte Loh, **Owen Dugan**, Di Luo, Marin Soljačić
QuanTA: Efficient High-Rank Fine-Tuning of LLMs with Quantum-Informed Tensor Adaptation
[ARXIV:2406.00132](#)
- 10) J. W. Liu, J. Grogan, **O. M. Dugan**, S. Arora, A. Rudra, C. Ré
Can Transformers Solve Least Squares to High Precision?
Workshop on Efficient Systems for Foundation Models II @ ICML 2024.
[ARXIV:2407.02405](#)
- 9) **Owen Dugan**
Machine Learning for Physics: from Symbolic Regression to Quantum Simulation
MIT Undergraduate Physics Thesis, 2024. [Thesis Link](#).
- 8) **Owen Dugan**, Peter Lu, Rumen Dangovski, Di Luo, Marin Soljačić
Q-Flow: Generative Modeling for Differential Equations of Open Quantum Dynamics with Normalizing Flows
Proceedings of the 40th International Conference on Machine Learning, Honolulu, Hawaii. PMLR 202, 2023.
[ARXIV:2302.12235](#)
- 7) **Owen Dugan**, Rumen Dangovski, Allan Costa, Samuel Kim, Pawan Goyal, Joseph Jacobson, Marin Soljačić
OccamNet: A Fast Neural Model for Symbolic Regression at Scale
[ARXIV:2007.10784](#)
- 6) Julia Balla, Sihao Huang, **Owen Dugan**, Rumen Dangovski, Marin Soljačić
AI-Assisted Discovery of Quantitative and Formal Models in Social Science
[ARXIV:2210.00563](#)
- 5) **Owen Dugan**
QiskiFT: Quantum Error Correction and Quantum Fault Tolerance Development Kit
[Documentation](#)
- 4) **Owen Dugan**
Astronomy Will Not Trail Off: Novel Methods for Removing Satellite Trails from Celestial Images
Journal of the American Association of Variable Star Observers, vol. 48, no. 2, p. 262, 2020. (Abstract only.)
- 3) Peyton Robertson, Connor Espenshade, Jay Sarva, **Owen Dugan**, Kalée Tock
An Automated Approach to Modeling Jupiter's Synchrotron Radiation from Radio Telescope Observations
Astronomy Theory, Observations and Methods, vol. 1, no. 1, pp. 24-33, 2020.
- 2) **Owen Dugan**, Thomas Robinson, Finnian Carmeci, Kalée Tock
CCD Measurements and Reclassification of WDS 07106 +1543 to an Optical Double
Journal of Double Star Observations, vol. 15, no. 1, pp. 119-129, 2019.
- 1) **Owen Dugan**, James Krasner
Soup, Bones, and Shakespeare: Literary Authorship and Allusion in Middle-earth
Mythlore, vol. 40, no. 2, pp. 105-120, 2022.

PRESENTATIONS

8) OccamLLM: Fast and Exact Language Model Arithmetic in a Single Step <i>The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024)</i>	December 2024
7) QuanTA: Efficient High-Rank Fine-Tuning of LLMs with Quantum-Informed Tensor Adaptation <i>The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024)</i>	December 2024
6) Q-Flow: Generative Modeling for Differential Equations of Open Quantum Dynamics with Normalizing Flows <i>American Physical Society (APS) March Meeting</i>	March 2024
5) Q-Flow: Generative Modeling for Differential Equations of Open Quantum Dynamics with Normalizing Flows <i>International Conference on Machine Learning</i>	July 2023
4) OccamNet: A Feed-Forward Neural Model for Symbolic Regression <i>MIT Conference on Mechanistic Interpretability</i>	May 2023
3) Q-Flow: Generative Modeling for Differential Equations of Open Quantum Dynamics with Normalizing Flows <i>Institute for AI and Fundamental Interactions – External Advisory Board Review</i>	May 2023
2) Q-Flow: Generative Modeling for Differential Equations of Open Quantum Dynamics with Normalizing Flows <i>Institute for AI and Fundamental Interactions – Mini Symposium</i>	April 2023
1) Astronomy Will Not Trail Off: Novel Methods for Removing Satellite Trails from Celestial Images <i>Joint Meeting of the Society for Astronomical Sciences (SAS) and the American Association of Variable Star Observers (AAVSO 108th Spring Meeting)</i>	May 2020

ADDITIONAL AWARDS & ACCOMPLISHMENTS

Sigma Pi Sigma Member , Sigma Pi Sigma Society	2024
Phi Beta Kappa Member , Phi Beta Kappa Society	2024
MIT Advanced Standing Examinations , Earned credit for 8 MIT classes	2021